

CNN Architecture and LeNet-5

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1. Overview

Key point: A CNN stacks blocks of convolution and pooling, flattens the result, and finishes with fully connected layers and an output layer. LeNet-5 (LeCun et al. 1998) is the classic example: two convolution-and-pooling blocks, then layers of 120, 84 and 10 nodes, about 60,000 parameters in all.

The previous Notes built the parts: the [convolution operation](#), [padding and strides](#) and [pooling](#). This Note puts them together into a whole network, first in general and then in the first CNN that solved a real problem.

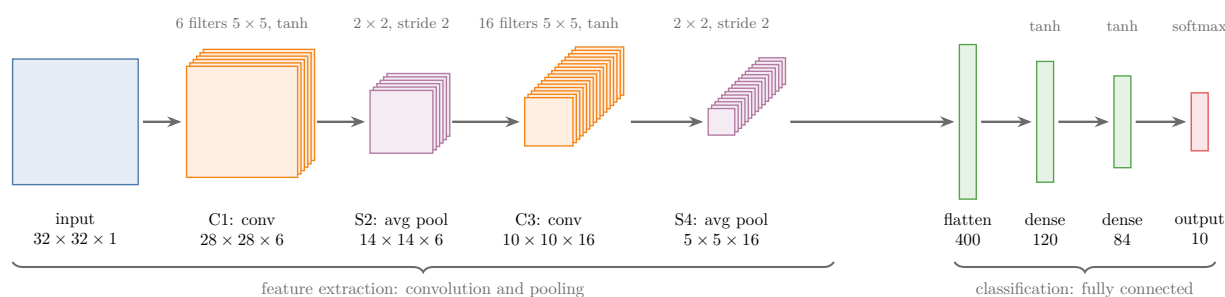


Figure 1: LeNet-5: two blocks of convolution (tanh) and average pooling turn a 32×32 image into 16 maps of 5×5 ; these are flattened to 400 numbers and passed through fully connected layers of 120, 84 and 10 nodes

Figure 1 shows LeNet-5. This Note covers the general CNN architecture, what can be varied in it, the LeNet-5 layers and their shapes, the parameter count, and LeNet-5 in Keras trained on MNIST.

2. Prerequisites

- The [convolution operation Note](#), the [padding and strides Note](#) and the [pooling Note](#).
- The [MNIST ANN Note](#): Flatten, Dense layers, softmax output and counting parameters.
- The [activation functions Note](#): tanh and ReLU.

3. The general CNN architecture

Key point: Input, then one or more blocks of convolution + ReLU + pooling, then Flatten, then any number of fully connected layers, then the output layer.

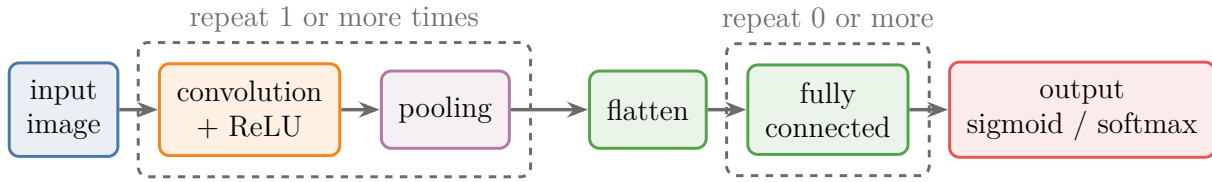


Figure 2: The common pattern of a CNN: convolution and pooling blocks extract features; flatten and fully connected layers classify

Almost every CNN follows the pattern of Figure 2:

1. **Input:** an image, for example an RGB image of $32 \times 32 \times 3$.
2. **Convolution layer:** a set of filters (kernels), say 3, each with 3 channels because the input has 3. The output is a volume of feature maps with 3 channels, one per filter.
3. **Non-linearity:** an activation function such as ReLU applied to every number of the feature maps.
4. **Pooling layer:** shrinks the volume.
5. **Repeat** steps 2–4 as many times as needed: a second convolution, a second pooling, and so on.
6. **Flatten:** turn the final 3D volume into one long 1D vector of numbers.
7. **Fully connected layers:** the Dense layers of an ANN, as many as the problem needs.
8. **Output layer:** one node with sigmoid for binary classification, or one node per class with softmax for multi-class classification.

The convolution and pooling blocks extract the features; the fully connected part uses them to classify. The CS231n notes write the same pattern as $\text{INPUT} \rightarrow [(\text{CONV} \rightarrow \text{RELU}) \times N \rightarrow \text{POOL}] \times M \rightarrow [\text{FC} \rightarrow \text{RELU}] \times K \rightarrow \text{FC}$.

3.1 What can be varied

Key point: The number of convolution layers and filters, filter size, stride, padding, the fully connected part, the activation, and extras such as dropout and batch normalisation.

Different CNN architectures come from different choices for:

- the number of convolution layers, and the number of filters in each;
- the filter size, the stride and whether there is padding;
- the number of fully connected layers and of nodes in each;
- the activation function;
- whether [dropout](#) or [batch normalisation](#) is used.

Over the years, competitions such as ImageNet, a hard image classification task with many classes, have produced a series of famous architectures: LeNet, AlexNet, GoogLeNet, VGGNet, ResNet and Inception. All of them follow the pattern above with different choices. LeNet came first.

4. LeNet-5

Key point: Yann LeCun’s network for reading handwritten digits: input 32×32 , convolution with 6 filters of 5×5 , average pooling, convolution with 16 filters of 5×5 , average pooling, then 120, 84 and 10 nodes, with tanh activations.

4.1 Background

Key point: LeCun worked on CNNs from 1989; the 1998 paper describes LeNet-5, used in a commercial system that read millions of cheques a day.

Yann LeCun, often called the father of CNNs, worked on convolutional networks from 1989. An early network of his read handwritten US Postal Service zip codes (LeCun et al. 1998 describe LeNet-1 as developed on this zip-code data). In 1998 he and his co-authors published LeNet-5 for handwritten digit recognition. A check-reading system built on CNN character recognisers “is deployed commercially and reads several million checks per day” (LeCun et al. 1998).

4.2 The layers

Key point: conv $6@5 \times 5 \rightarrow$ avg pool $2 \times 2 \rightarrow$ conv $16@5 \times 5 \rightarrow$ avg pool $2 \times 2 \rightarrow$ flatten $\rightarrow 120 \rightarrow 84 \rightarrow 10$.

LeNet-5 expects a 32×32 greyscale image. Layer by layer (Figure 1):

1. **Convolution 1:** 6 filters of 5×5 , stride 1, no padding, tanh.
2. **Average pooling 1:** 2×2 window, stride 2. LeNet-5 uses average pooling, not max pooling.
3. **Convolution 2:** 16 filters of 5×5 , stride 1, no padding, tanh.
4. **Average pooling 2:** 2×2 window, stride 2.
5. **Flatten.**
6. **Fully connected:** 120 nodes, tanh.
7. **Fully connected:** 84 nodes, tanh.
8. **Output:** 10 nodes with softmax, one for each digit.

The activation is **tanh**, not ReLU. ReLU is the usual choice in today’s CNNs, but in 1998 tanh was the best activation function available (see the [activation functions Note](#)).

4.3 The shapes, layer by layer

Key point: $32 \rightarrow 28 \rightarrow 14 \rightarrow 10 \rightarrow 5$, then $5 \times 5 \times 16 = 400$ numbers.

The sizes follow from the formulas of the [padding and strides Note](#): $n - f + 1$ for a convolution without padding, $\lfloor (n - 2)/2 \rfloor + 1$ for 2×2 pooling with stride 2.

Layer	Calculation	Output
Input		$32 \times 32 \times 1$
Convolution 1, 6 filters 5×5	$32 - 5 + 1 = 28$	$28 \times 28 \times 6$
Average pooling 1	$28/2 = 14$	$14 \times 14 \times 6$
Convolution 2, 16 filters 5×5	$14 - 5 + 1 = 10$	$10 \times 10 \times 16$
Average pooling 2	$10/2 = 5$	$5 \times 5 \times 16$
Flatten	$5 \times 5 \times 16$	400
Dense		120
Dense		84
Output		10

So the first fully connected layer has 400×120 weights, the second 120×84 , and the output layer 84×10 .

The number of filters grows as we go deeper, from 6 to 16, while the filter size stays 5×5 and the height and width of the maps shrink. More filters in deeper layers is a pattern we will see again in later architectures.

4.4 Why “5”

Key point: The paper calls LeNet-5 a network of 7 layers, not counting the input; “5” is the version number after LeNet-1 and LeNet-4.

One common way to count reaches 5 layers: each convolution with its pooling counts as one layer, giving 2, plus the two hidden fully connected layers and the output layer. But the paper itself says “LeNet-5 comprises 7 layers, not counting the input”: C1, S2, C3, S4, C5, F6 and the output (LeCun et al. 1998). The same paper also tests LeNet-1 and LeNet-4, so the 5 is best read as a version number.

5. Counting the parameters

Key point: A convolution layer has $(\text{filter height} \times \text{width} \times \text{input channels} + 1) \times \text{filters parameters}$. LeNet-5 in Keras has 61,706, and 48,120 of them sit in the first fully connected layer.

1. **In words:** each filter has $\text{height} \times \text{width} \times \text{input channels}$ weights plus one bias; a Dense layer has $\text{inputs} \times \text{nodes}$ weights plus one bias per node. Pooling and Flatten have none.
2. **Formula:**

$$\text{conv parameters} = (f \times f \times c_{\text{in}} + 1) \times k, \quad \text{dense parameters} = (n_{\text{in}} + 1) \times n_{\text{out}}$$

3. **Example:** convolution 1 has $(5 \times 5 \times 1 + 1) \times 6 = 156$ parameters; convolution 2 has $(5 \times 5 \times 6 + 1) \times 16 = 2,416$.

From `model.summary()` in the Notebook:

Layer	Output shape	Parameters
Conv2D, 6 filters 5×5	$28 \times 28 \times 6$	156
AveragePooling2D	$14 \times 14 \times 6$	0
Conv2D, 16 filters 5×5	$10 \times 10 \times 16$	2,416
AveragePooling2D	$5 \times 5 \times 16$	0
Flatten	400	0
Dense, 120	120	48,120
Dense, 84	84	10,164
Dense, 10	10	850
Total		61,706

Pooling layers have no parameters because pooling involves no training, and Flatten has none because it only reshapes. LeNet-5 has about 60,000 parameters, few for a CNN; modern CNNs have millions.

Extra: The original LeNet-5 differs from the Keras version in a few details, all from LeCun et al. (1998). Its pooling layers multiply each average by a trainable coefficient and add a trainable bias (12 and 32 parameters). C3 connects each of its 16 maps to only some of S2's 6 maps (1,516 parameters instead of 2,416). C5 is a convolution layer with 120 filters of 5×5 , which on a 5×5 input amounts to a fully connected layer. The activation is a scaled tanh, $1.7159 \tanh(Sa)$, and the output layer uses Euclidean radial basis function units instead of softmax. Adding the paper's own counts, $156 + 12 + 1,516 + 32 + 48,120 + 10,164 = 60,000$, the "60,000 trainable free parameters" the paper states.

6. LeNet-5 in Keras

Key point: Two Conv2D + AveragePooling2D blocks, Flatten, three Dense layers. On MNIST padded to 32×32 it reaches 98.46% test accuracy after 10 epochs.

Python: LeNet-5 with today's Keras layers.

```
model = keras.Sequential([
    keras.Input(shape=(32, 32, 1)),
    keras.layers.Conv2D(6, kernel_size=(5, 5), strides=1,
                        padding="valid", activation="tanh"),
    keras.layers.AveragePooling2D(pool_size=(2, 2), strides=2),
    keras.layers.Conv2D(16, kernel_size=(5, 5), strides=1,
                        padding="valid", activation="tanh"),
    keras.layers.AveragePooling2D(pool_size=(2, 2), strides=2),
    keras.layers.Flatten(),
    keras.layers.Dense(120, activation="tanh"),
    keras.layers.Dense(84, activation="tanh"),
```

```
keras.layers.Dense(10, activation="softmax"]])
```

MNIST digits are 28×28 , so the Notebook adds 2 rows and columns of zeros on every side to make them 32×32 . The paper's input was also larger than the digits, so that features at the edge of a digit can sit in the centre of a filter's view (LeCun et al. 1998). Pixels are scaled to 0–1, and the model is compiled as in the [MNIST ANN Note](#) (sparse categorical cross-entropy, Adam).

Trained for 10 epochs with batch size 128, averaged over 3 seeds, LeNet-5 reaches a test accuracy of 98.46%, a test error of 1.54%. The paper reports 0.95% test error for LeNet-5 (LeCun et al. 1998), trained differently and for longer. The ANN of the [MNIST ANN Note](#) reached 97.63%.

7. Summary

	General CNN	LeNet-5
Feature extraction	$[\text{conv} + \text{ReLU} \rightarrow \text{pool}] \times M$	$2 \times [\text{conv } 5 \times 5 \text{ (6, then 16 filters)} + \text{tanh} \rightarrow \text{avg pool } 2 \times 2]$
Bridge	Flatten	$5 \times 5 \times 16 \rightarrow 400$
Classification	fully connected $\times K$	$120 \rightarrow 84 \text{ (tanh)}$
Output	sigmoid or softmax	10, softmax
Parameters	depends	61,706 in Keras (60,000 in the paper)

- A CNN = convolution and pooling blocks, Flatten, fully connected layers, output layer.
- Architectures differ in the number of layers and filters, filter size, stride, padding, activation, dropout and batch normalisation.
- LeNet-5 (LeCun et al. 1998): 32×32 input, tanh, average pooling, 6 then 16 filters of 5×5 , layers of 120, 84 and 10 nodes.
- Pooling and Flatten have no parameters; most of LeNet-5's parameters are in its first Dense layer.

8. Sources

- LeCun, Y., Bottou, L., Bengio, Y. and Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278–2324. Section II.B (LeNet-5), Table I, Section III.
- Keras API documentation: `Conv2D`, `AveragePooling2D`, `Flatten` and `Dense` layers, keras.io/api/layers.
- Stanford CS231n course notes, “Convolutional Neural Networks”, section “Layer Patterns”, cs231n.github.io/convolutional-networks.

9. Key terms

Term	Meaning
CNN architecture	The sequence of layers of a CNN: convolution and pooling blocks, Flatten, fully connected layers, output
Flatten layer	A layer that turns a volume into a 1D vector; no parameters
Fully connected (Dense) layer	A layer in which every node is connected to every input
LeNet-5	LeCun et al.'s 1998 CNN for handwritten digits: two conv-pool blocks, then 120, 84 and 10 nodes

Term	Meaning
Average pooling	Pooling that keeps the mean of each window; used in LeNet-5
ImageNet	A large image classification benchmark and competition that produced many CNN architectures